Embedded Control for Robotics

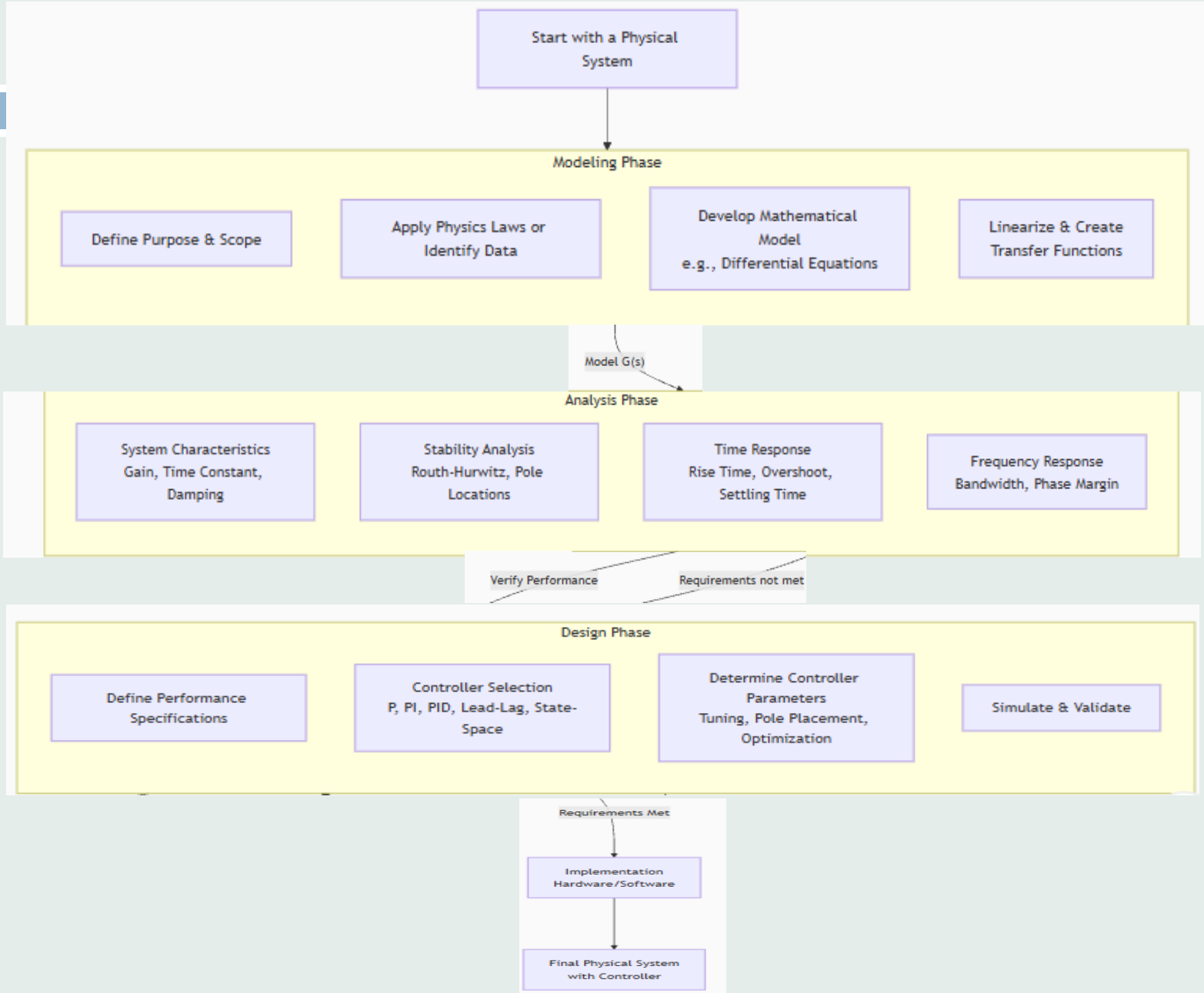
Lecture # 05 **TF, Pole/Zero, Stability, and Time Response**

- Book: Classical and Modern Control with Microcontrollers, Ch 2, 4

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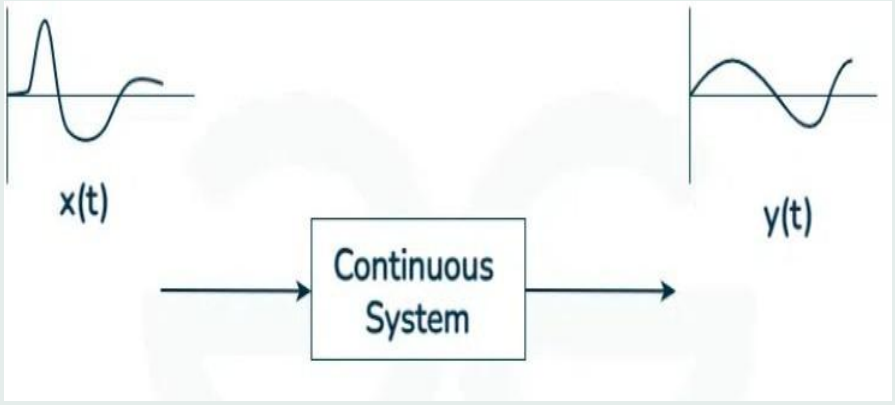
Control Systems: Big Picture

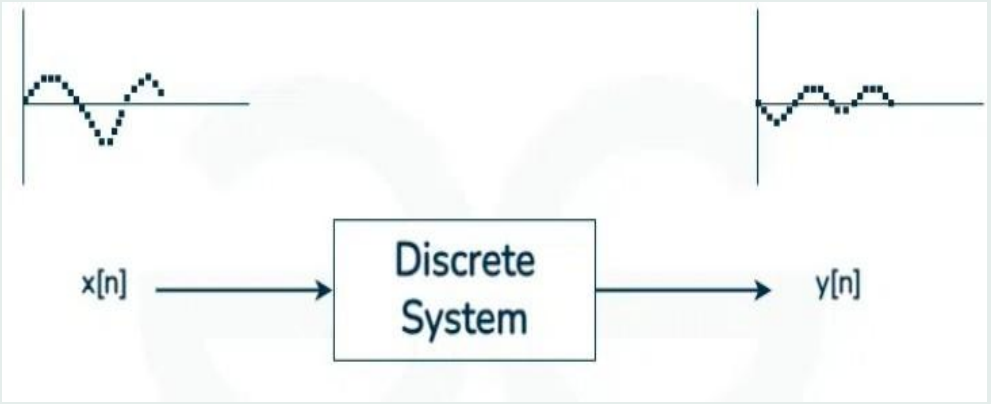
1. Modeling
2. Analysis
3. Design



Control Systems: Simplified Picture

Continuous Time		Discrete Time	
Differential Equations		Difference Equations	
Laplace Transform, Transfer Function	State Space Representation	Z transform, Transfer Function	State Space Representation
Root Locus Analysis	State Space: Modeling, Analysis, Design	Root Locus Analysis	State Space: Modeling, Analysis, Design



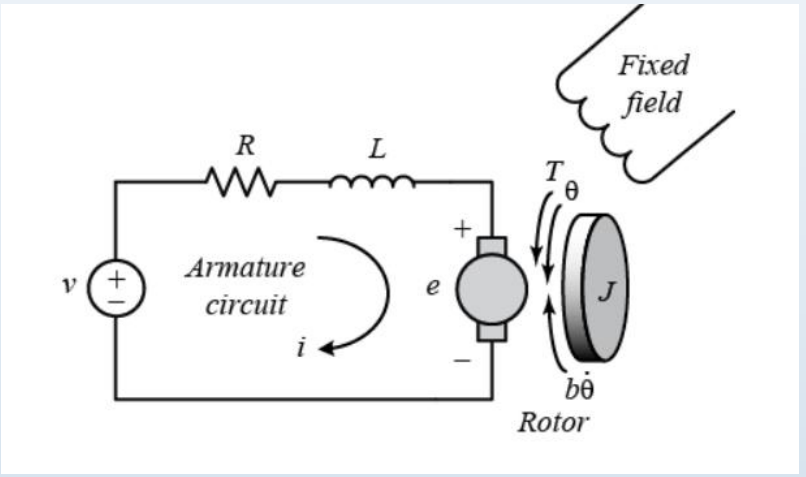


Overview Today's Lecture

- Transfer Function
- Pole/Zero and Stability
- Time Response of control System

DC Motor Speed: System Modeling

- DC motor provides rotary motion to the wheels of a robot.
- Model/Equation
- The torque generated by a DC motor is proportional to the armature current
- The back emf, e, is proportional to the angular velocity of the shaft
- Newton's 2nd law and Kirchhoff's voltage law.



$$e = K_e\dot{\theta}$$

$$T = K_t i$$

$$J\ddot{\theta} + b\dot{\theta} = Ki$$

$$L\frac{di}{dt} + Ri = V - K\dot{\theta}$$

Continue…

- Taking Laplace TF

$$s(Js+b)\,\theta(s) = KI(s)$$

$$(Ls+R)I(s) = V_i(s) - Ks\theta(s)$$

Combining Eqs. (4.13) and (4.14) to eliminate the intermediate variable $I(s)$, we get the transfer function for this DC motor system as

$$G(s) = \frac{\Omega(s)}{V_i(s)} = \frac{K}{(Js+b)(Ls+R)+K^2} \quad \left(\frac{\text{rad/s}}{\text{V}}\right) \tag{4.15}$$

where $\dot{\theta}(s) = s\theta(s) = \Omega(s)$ and it is the rotation speed of the motor (ω) measured by rad/s.

This is a second-order system with two poles and no zeros, and it is a typical transfer function used for most linear systems.

Analysis of Transfer Function

1. **Pole/Zero**
   - Why??
     - Pole/Zero Stability
2. **Time dynamic analysis**
     - DC Gain via step response
     - Time constant via step response
3. **Frequency Analysis**
     - Phase Shift
     - Amplitudes

•$G(s) = G(j\omega)$, ω is the frequency

The absolute value of s, $|s| = \sqrt{a^2 + b^2}$

•$|s|$ is often called ω_0

Characteristic Equation

set the denominator of the transfer function to zero to obtain the characteristic equation

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}$$

$$s^n + a_{n-1} s^{n-1} + \ + a_1 s + a_0 = 0$$

It n roots which determines the stability of the system

Poles and Zeros

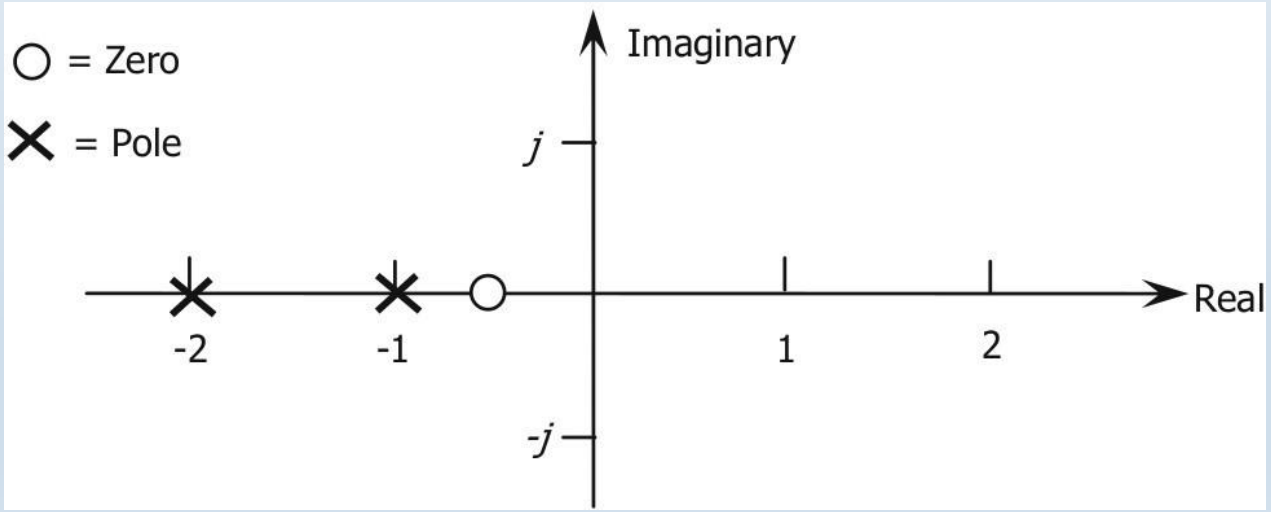
- The s-value to make the numerator equal to zero is called zeros
- The s-values to make denominator equal to zero are called poles
- Locations of the poles in a transfer function determine the properties for a control system

Pole-Zero form of TF

$$\frac{C(s)}{R(s)} = \frac{K(s + z_1)(s + z_2)\cdots(s + z_m)}{(s + p_1)(s + p_2)\cdots(s + p_n)}$$

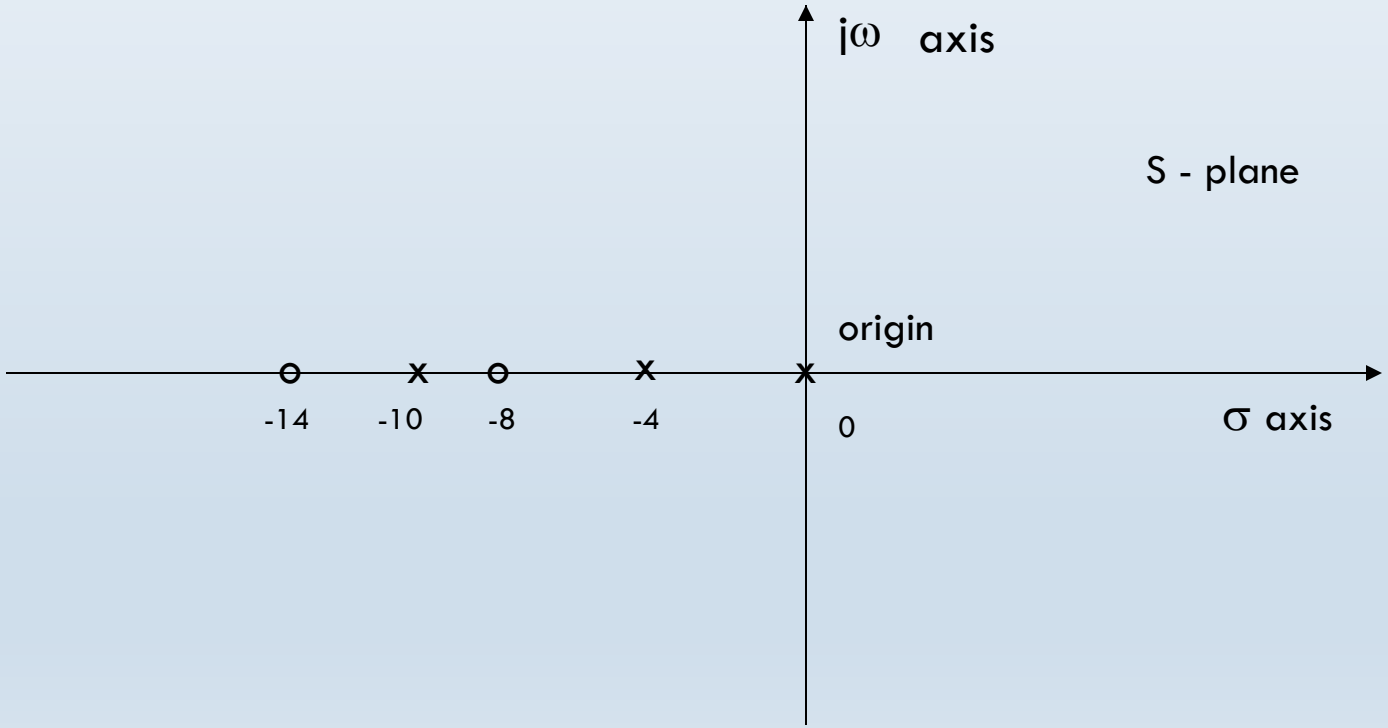
Poles and Zeros

$$G(s) = \frac{2s+1}{s^2+3s+2} = \frac{2(s+0.5)}{(s+1)(s+2)}$$



Poles and Zeros

$$G(s)=\frac{(s+8)(s+14)}{s(s+4)(s+10)}$$

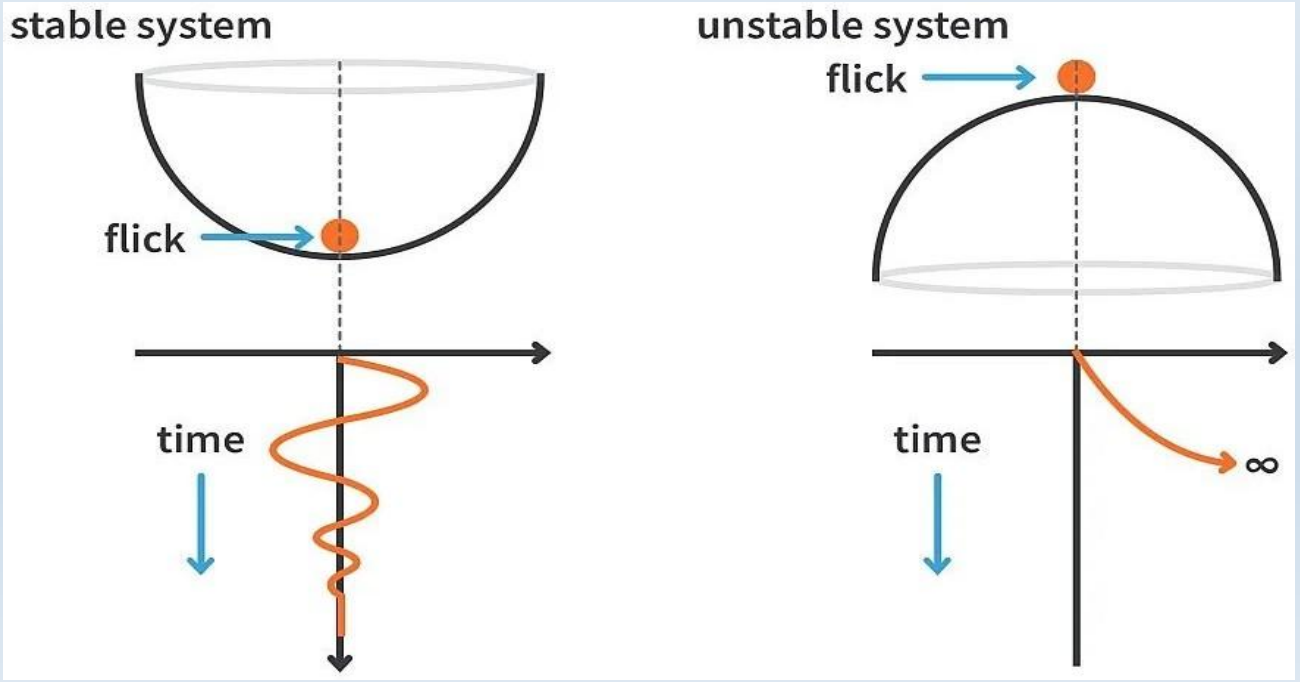


Stability

Key Concept: how a system responds to disturbances?

- A stable system absorbs disturbances and returns to equilibrium
- Unstable system amplifies them, leading to failure or chaos.

The image visually captures this by comparing the time response of both systems after a flick.



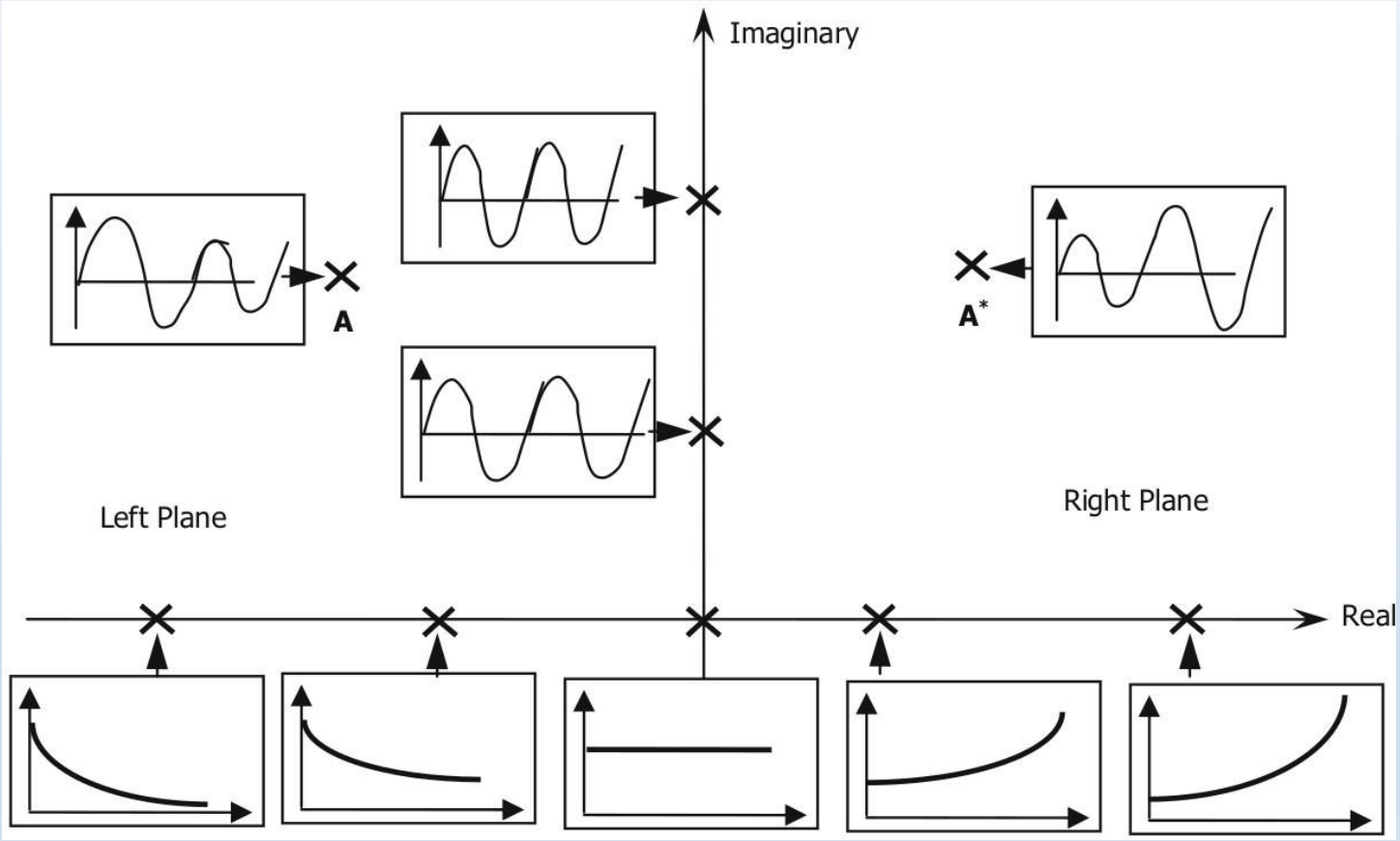
Example

- Real-world examples:

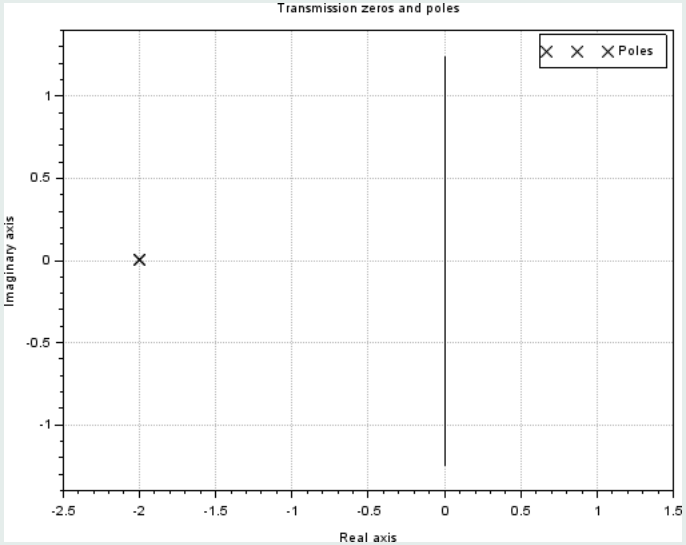
1. A pendulum that swings back to rest
2. a spring that returns to its natural length after being stretched.

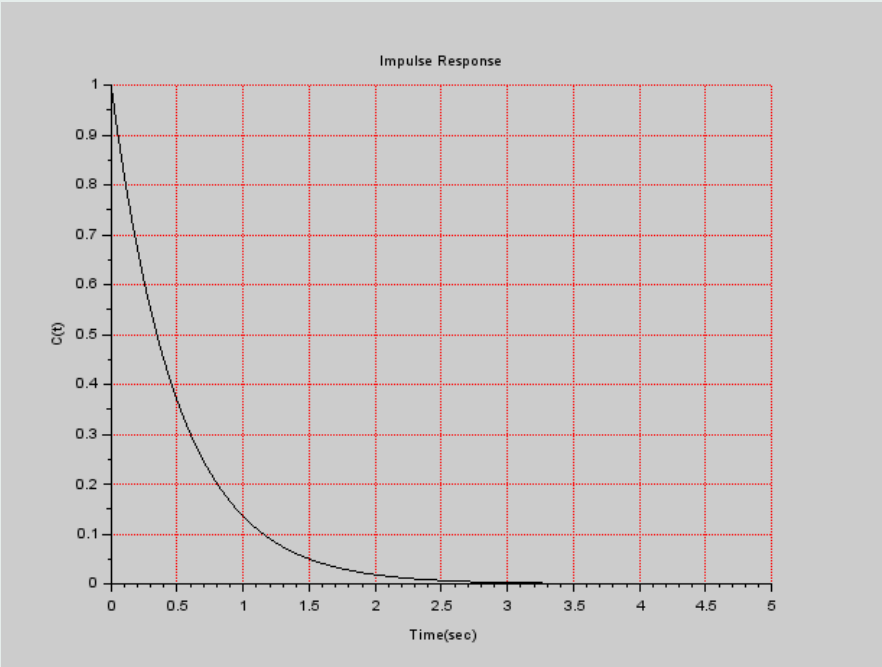
1. A pencil balanced on its tip that falls when nudged
2. feedback in a microphone causing a screech.

Location of Poles and Stability

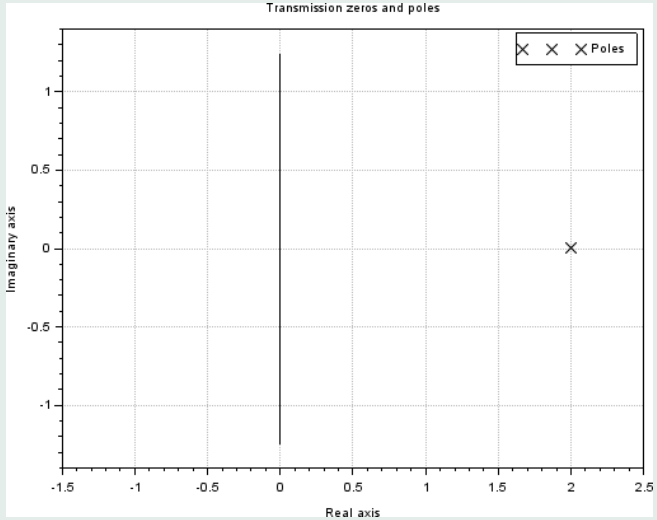


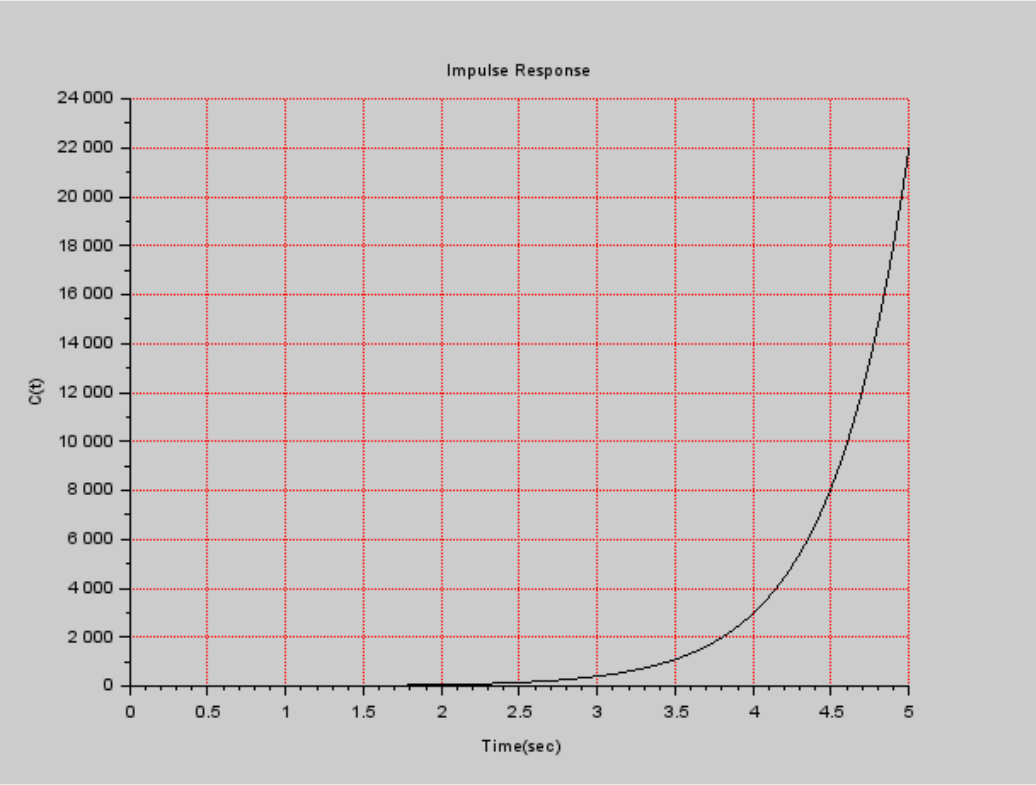
Case 1: Poles on the -ve real axis G(s) = 1/(s+2)



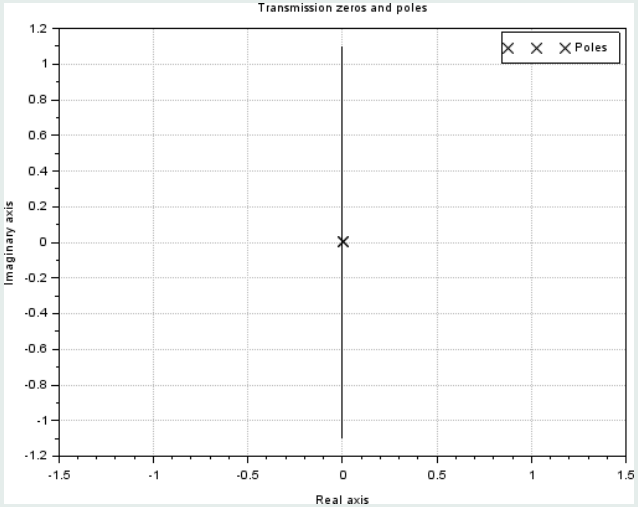


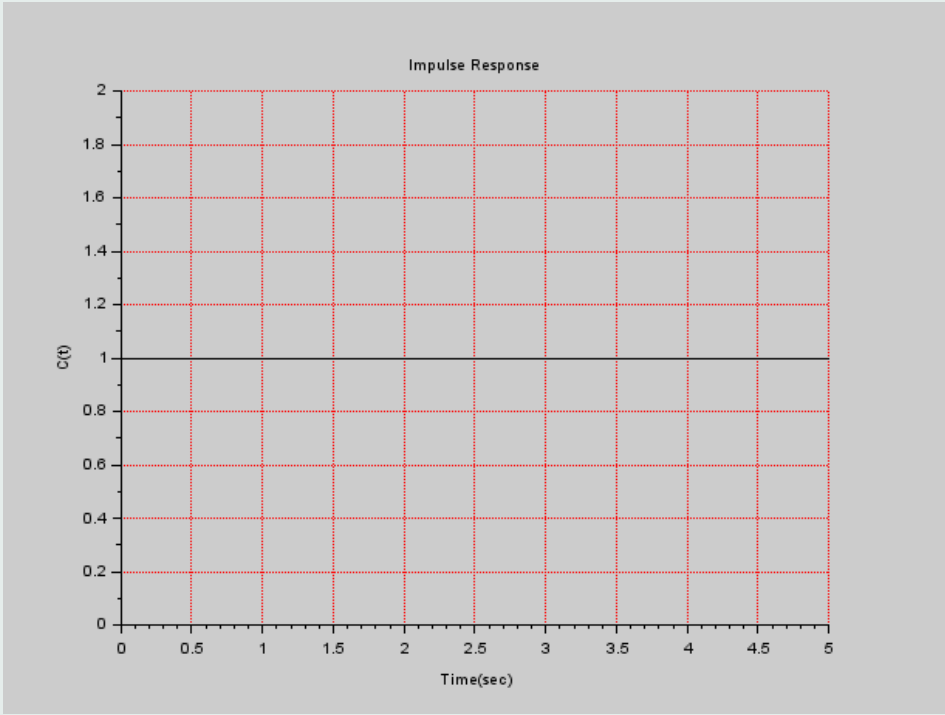
Case 2: Poles on positive real axis G(s) = 1/(s-2)



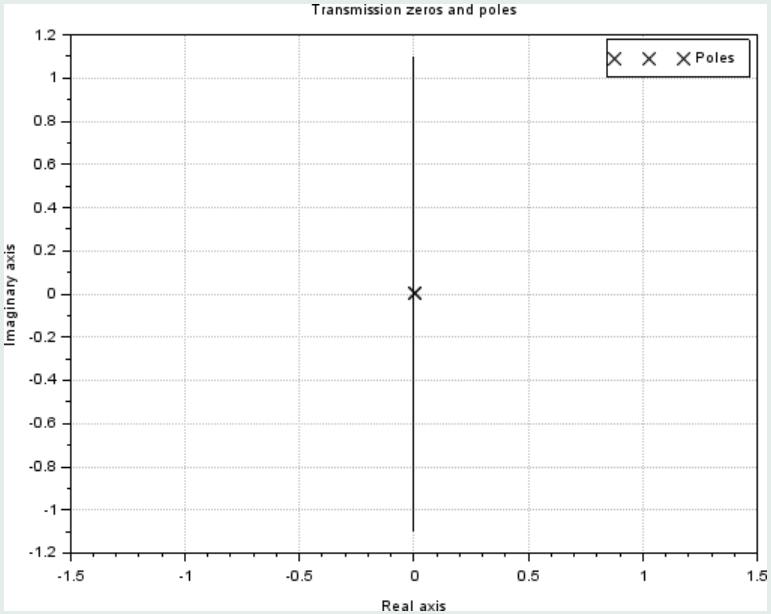


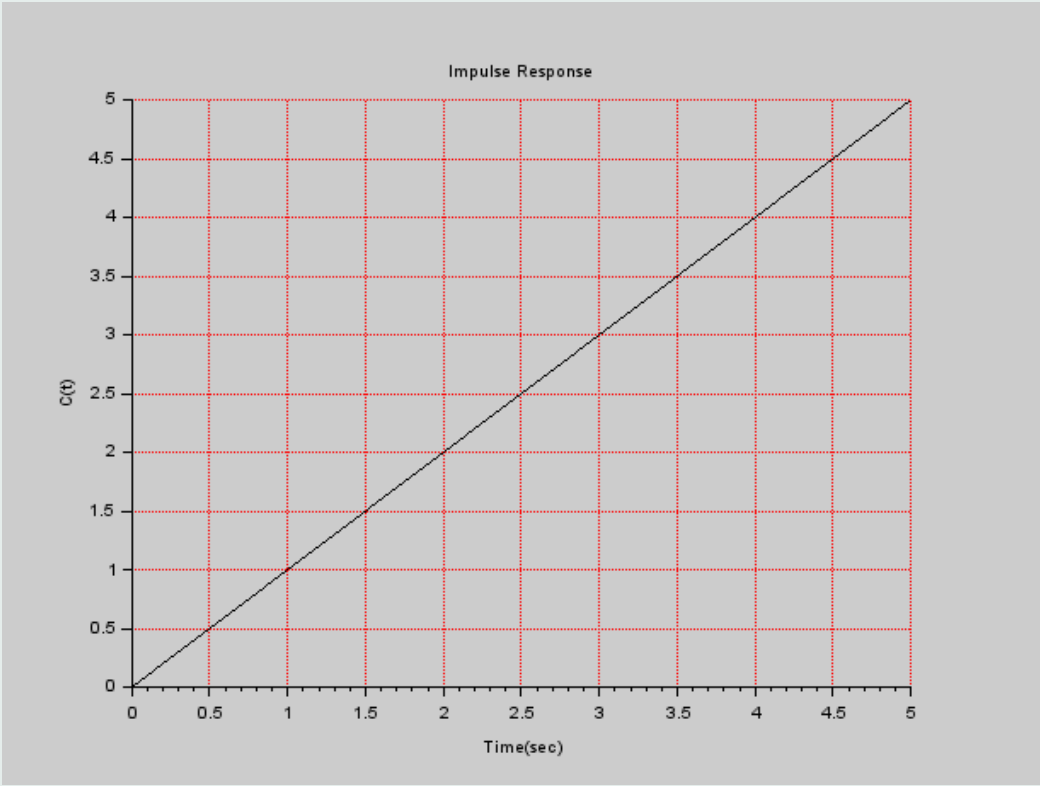
Case 3 – Poles at origin G(s) = 1/(s)



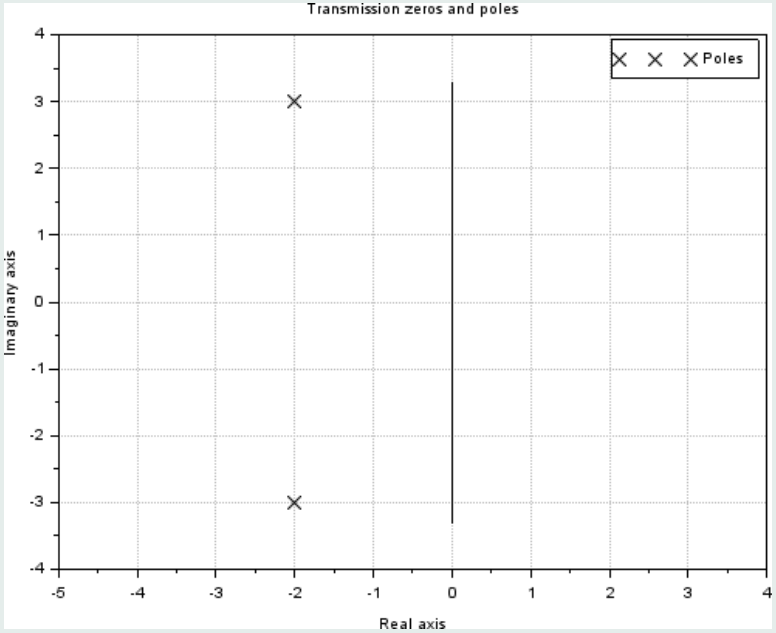


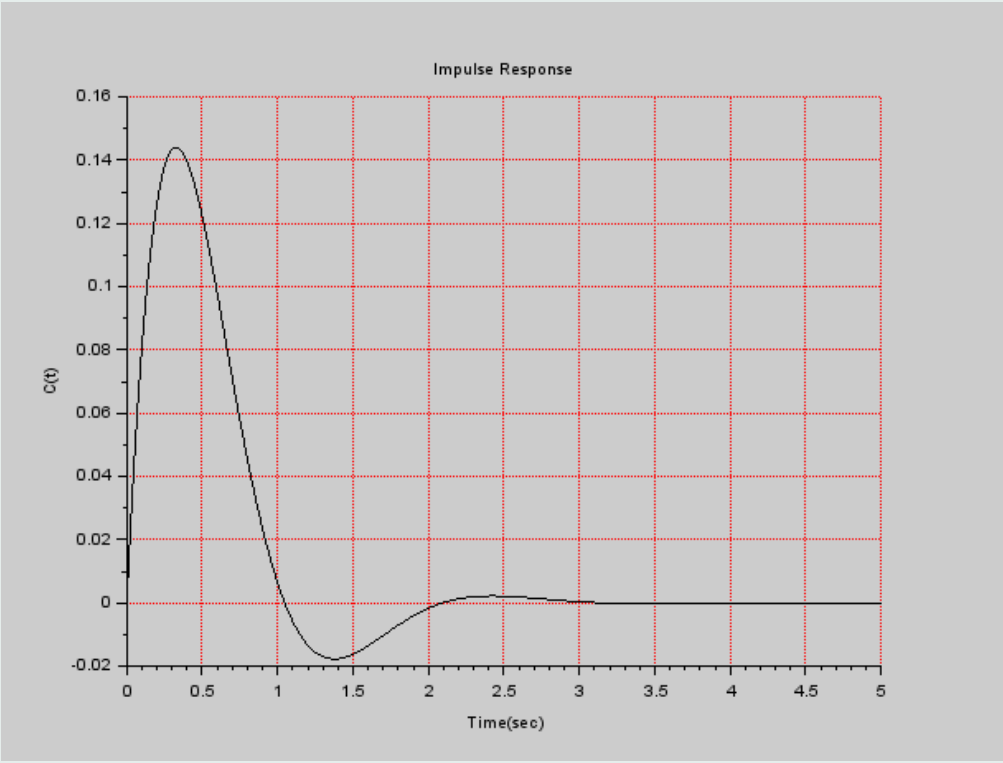
Case 3 – Poles at origin G(s) = 1/(s^2)



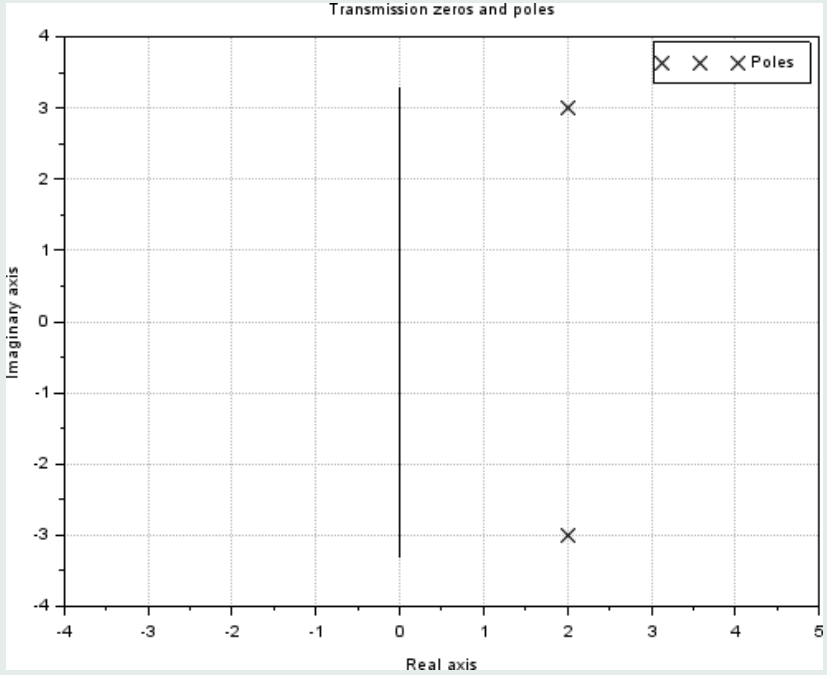


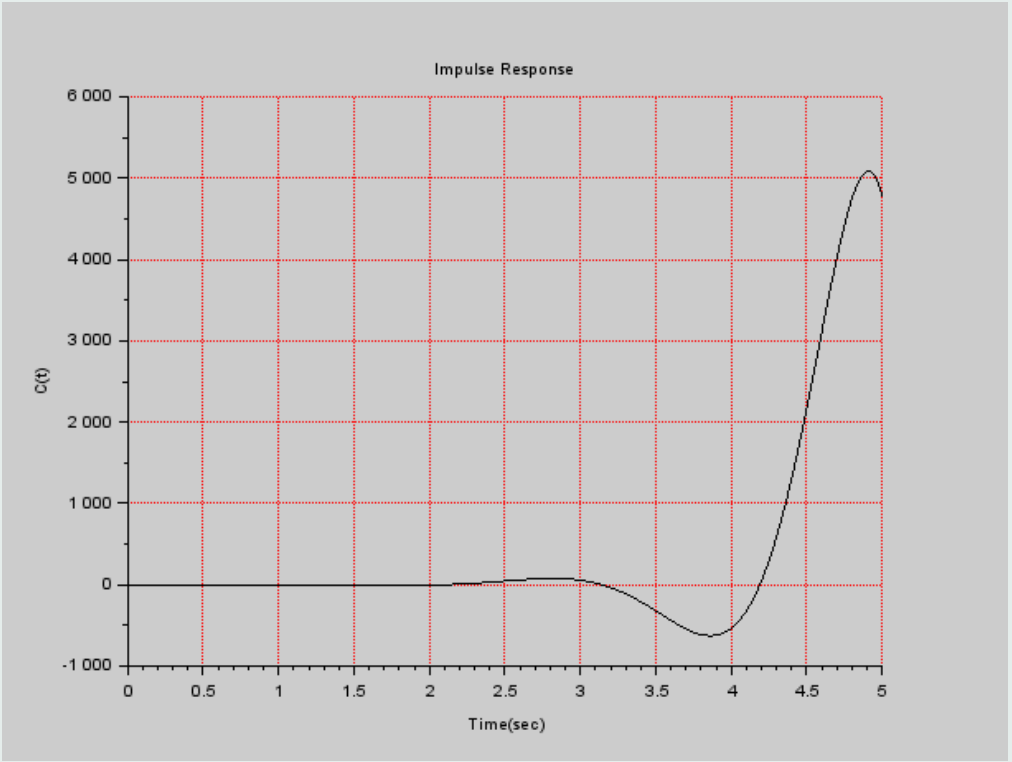
Case 4 – Complex pole in the left half of s-plane G(s) = 1/[(s+2-3j)(s+2+3j)]



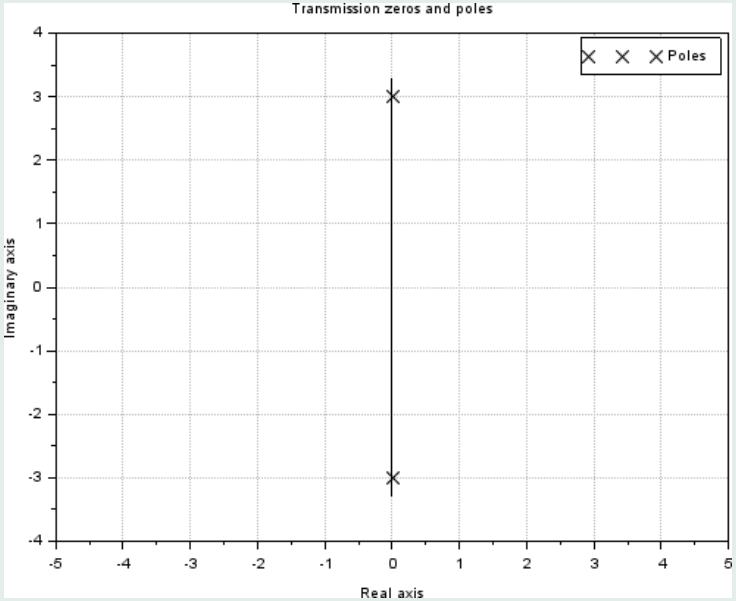


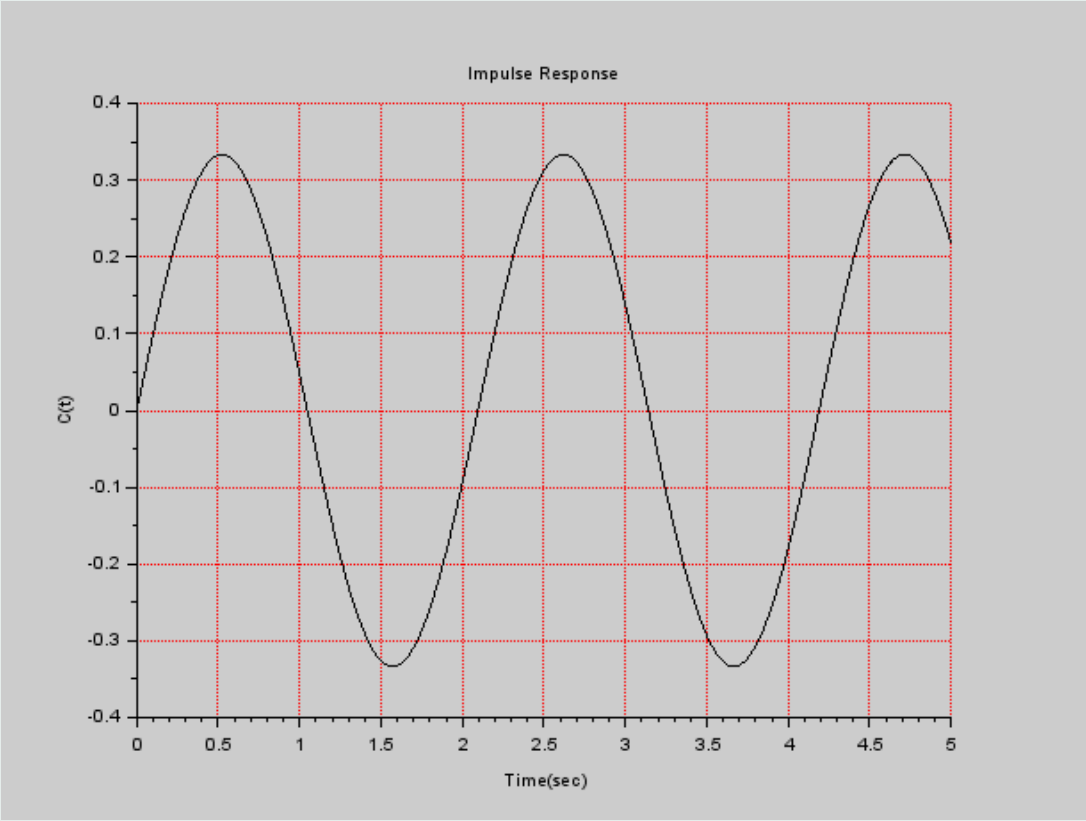
Case 5: Complex poles in right half of the s-plane G(s) = 1/[(s-2-3j)(s-2+3j)]





Case 6 – Poles on the imaginary axis G(s) = 1/[(s-3j)(s+3j)]





Summary: Pole Location & Stability

- If all the poles lie in the left half of the s-plane, then the system is stable.
- If any pole lies in the right half of the s-plane, then the system is unstable
- If the system has two or more poles in the same location on the imaginary axis, then the system is unstable.
- If the system has one or more non-repeated poles on the imaginary axis, then the system is marginally stable.

QUIZ: Draw Pole / Zero Plot

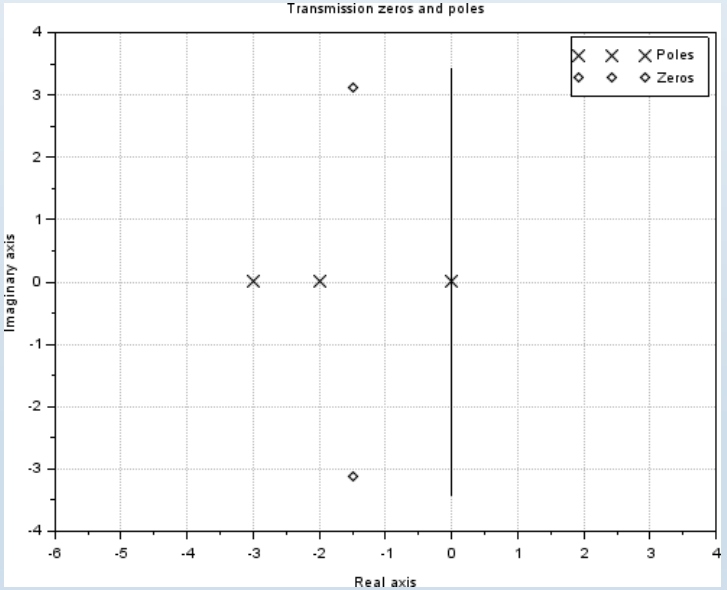
$$G(s) = \frac{s^2 + 3s + 12}{s(s^2 + 5s + 6)}$$

Continue…

$$G(s) = \frac{s^2 + 3s + 12}{s(s^2 + 5s + 6)}$$

Zeros as -1.5 ± 3.1225j

Poles as 0, -2, -3



Discussion

- Demo Python